MATH329 · OPERATIONS RESEARCH

Midterm 462 Practice

09/03/2025 · 30 marks

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01 Questions

Answer each question in the space below it. The full solution is in a separate file.

Q1 THEORY · 5 M

Answer the following:

a) When solving a linear programming problem graphically, there are special cases that we encounter, mention them?

b) To start the simplex algorithm, there are two requirements that must be met. What are they?

Q2 GRAPHICAL · 7 M

Solve the following problem graphically:

Maximum $z = 3x_1 + 2x_2$

S.to: $2x_1 + x_2 \leq 4$

$-2x_1 + x_2 \leq 2$

$x_2 \geq 1$

$x_1 \geq 0, x_2 \geq 0$

Q3 STANDARD FORM · 8 M

Given the following problem:

Maximum $Z = 8x_1 - 2x_2$

S.to $2x_1 + x_2 \leq 12$

$x_1 - 5x_2 \geq 6$

$4x_2 = -7$

$x_1 \geq 0$, x_2 unrestricted in sign.

1. What methods can be applied to solve the given linear programming problem?
2. Write the problem in standard form (equation form). "Don't solve the problem"

Q4 STANDARD FORM + SIMPLEX · 10 M

Given the following problem:

Min $Z = x_1 - 2x_2$

S.to $2x_1 + 3x_3 \leq 1$

$3x_1 + 2x_2 - x_3 \leq 5$

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$

a) Write the problem in standard form (equation form).

b) Solve the LP problem using the Simplex method.

Basic	x_1	x_2	x_3	s_1	s_2	Sol.
Z						
Z						
Z						